

Digital Signal Processing - Lecture 5

Spectrograms, the short-time Fourier transform, and windowing

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| Overview

This lecture

01 Why spectrograms?

02 How the STFT works

03 Windowing

04 Exercises

| Why spectrograms?

Listen to the two sounds

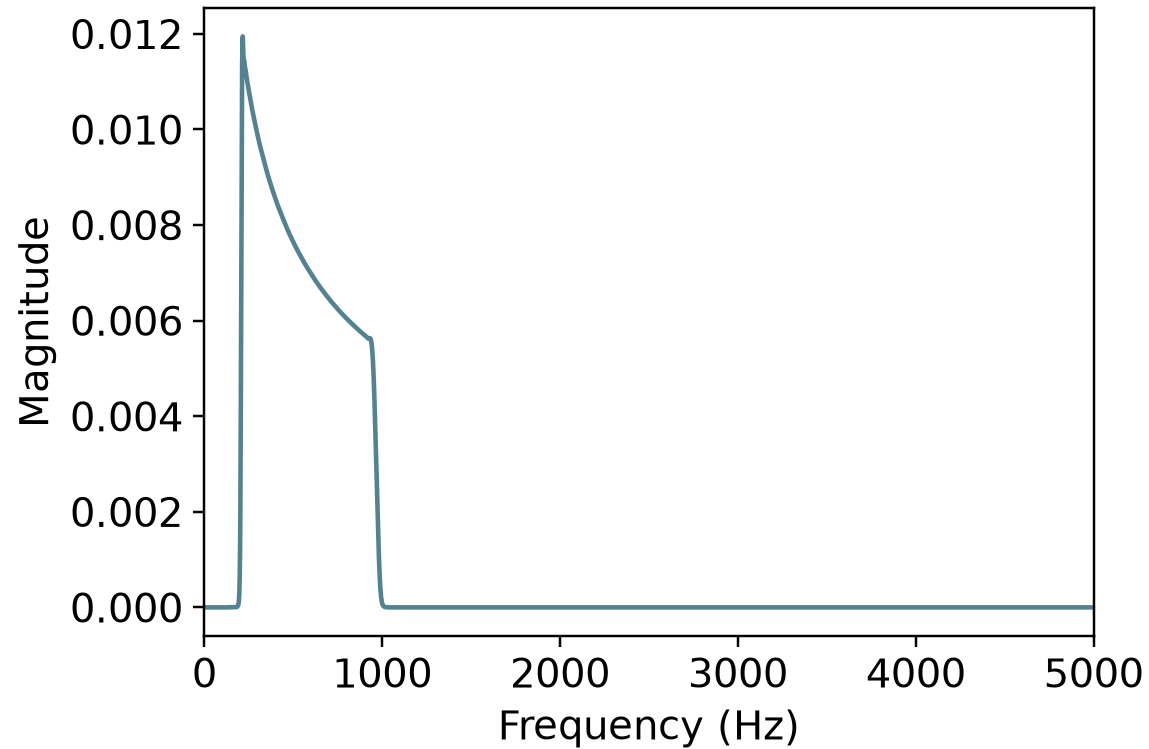
Sound A

Sound B

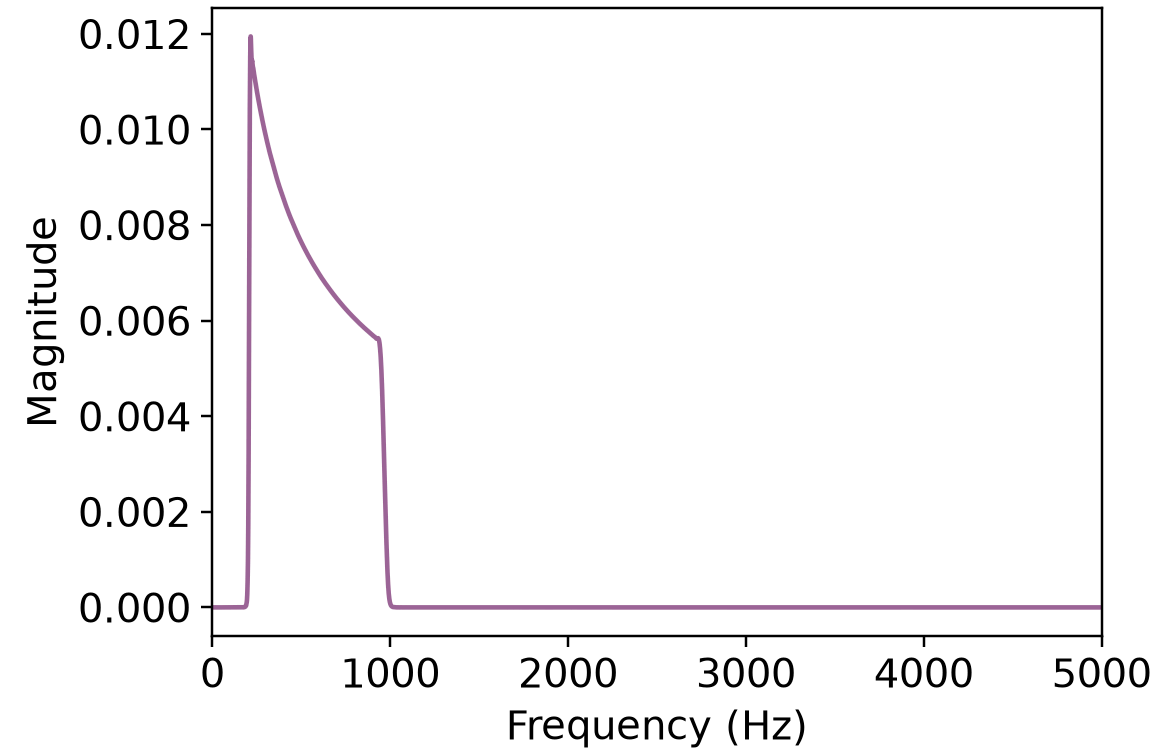
How does the pitch change in each sound?

Can the spectrum distinguish them?

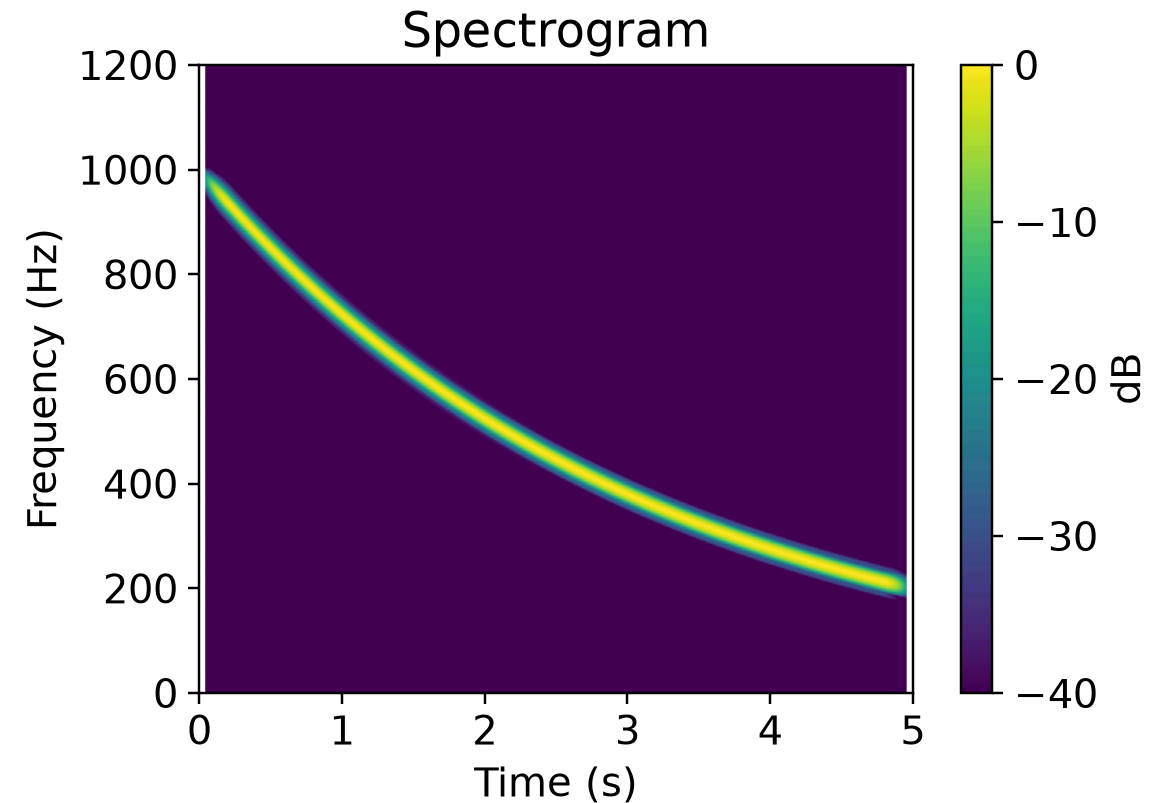
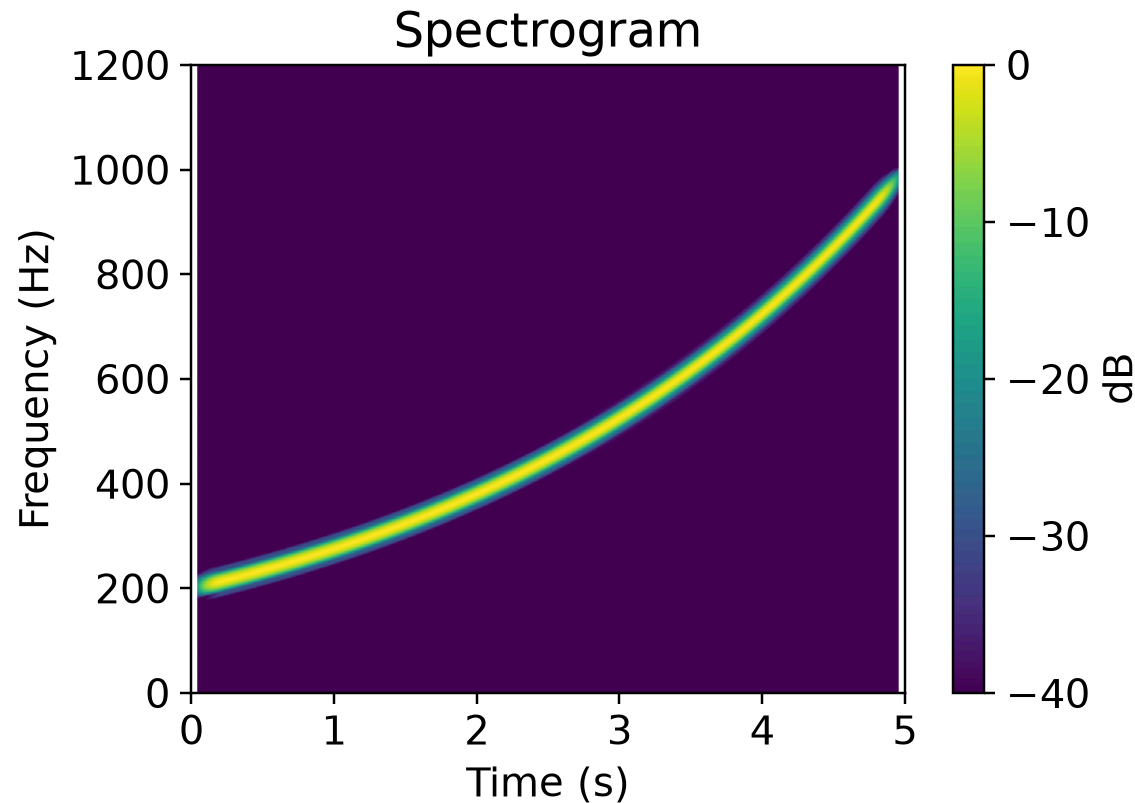
Spectrum plot



Spectrum plot



Frequency content over time



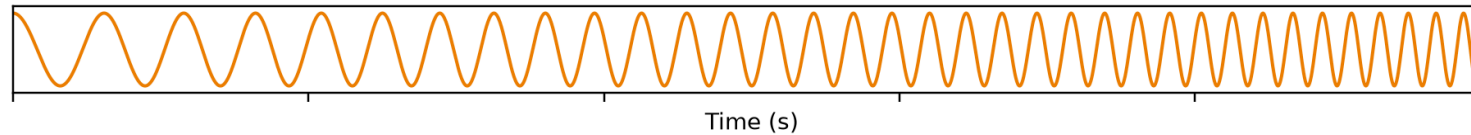
Time in horizontal axis, frequency in vertical axis. Brighter colours show stronger frequency components.

Short-time Fourier transform

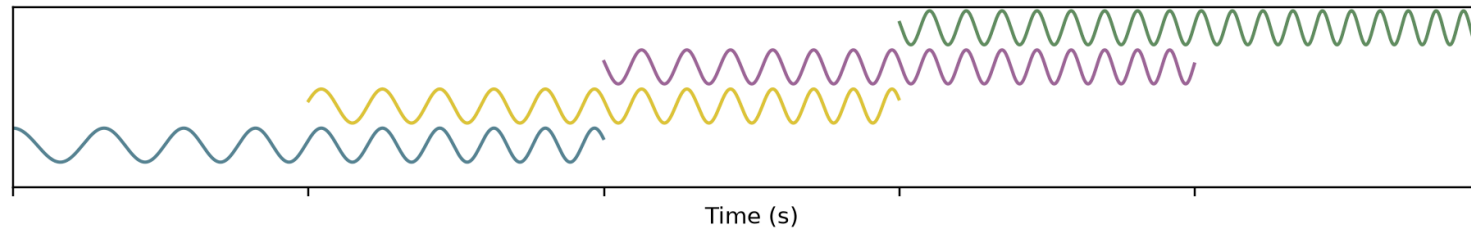
The spectra behind a spectrogram

From input signal to spectrogram

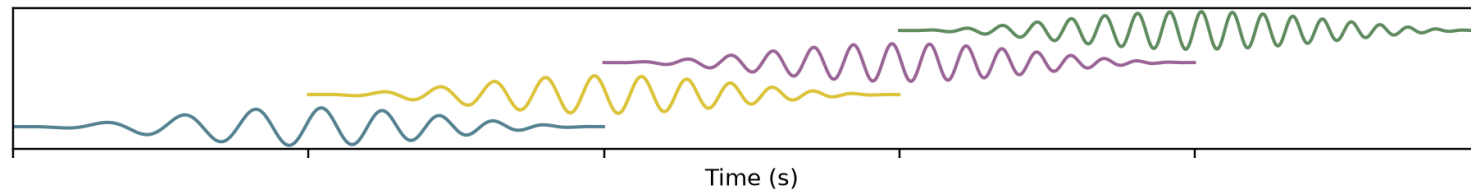
Input Signal
Time-domain



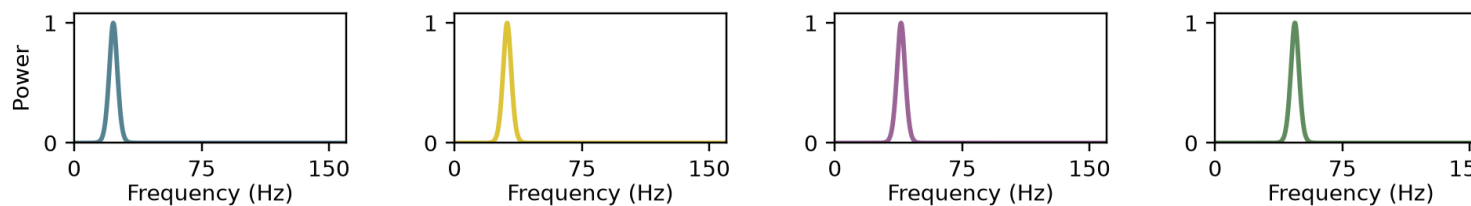
Segments
With overlap



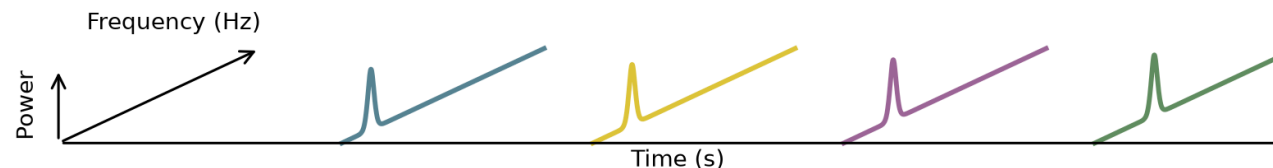
Apply window
Per segment, to
spectral leakage



FFT
Per windowed
segment



**Spectrogram
view**



Select overlapping segments

Amount of overlap is a configurable parameter. This stays in the time domain.

Apply a window

Multiply samples by window weights. Tapering the ends reduces spectral leakage as the ends are where partial periods occur. Still in the time domain.

Compute Fourier Transforms

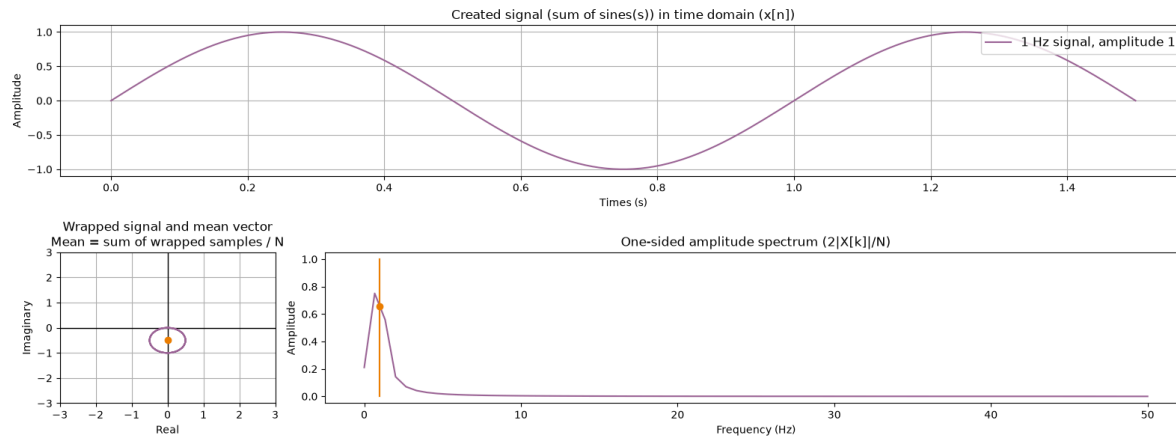
Apply the FFT to each windowed segment to obtain its frequency content. This converts the time-domain signal into its frequency-domain representation.

Stack the spectra in time

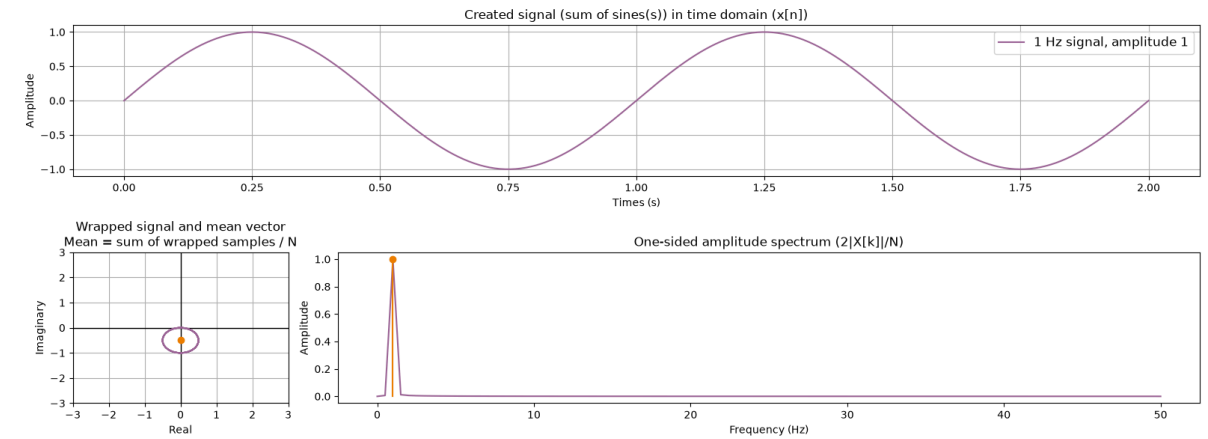
The height plotted here is typically colour coded in the spectrogram to represent the power at each frequency.

Recall spectral leakage

One and a half periods

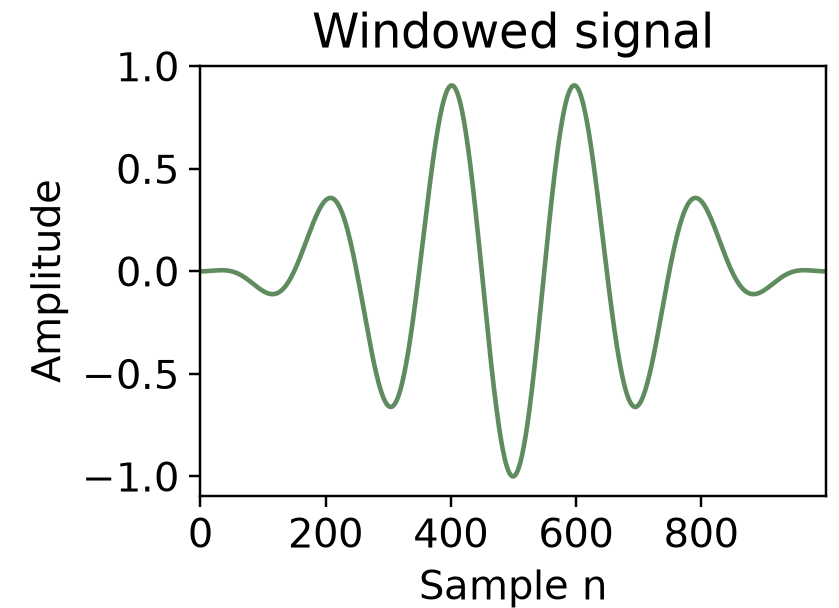
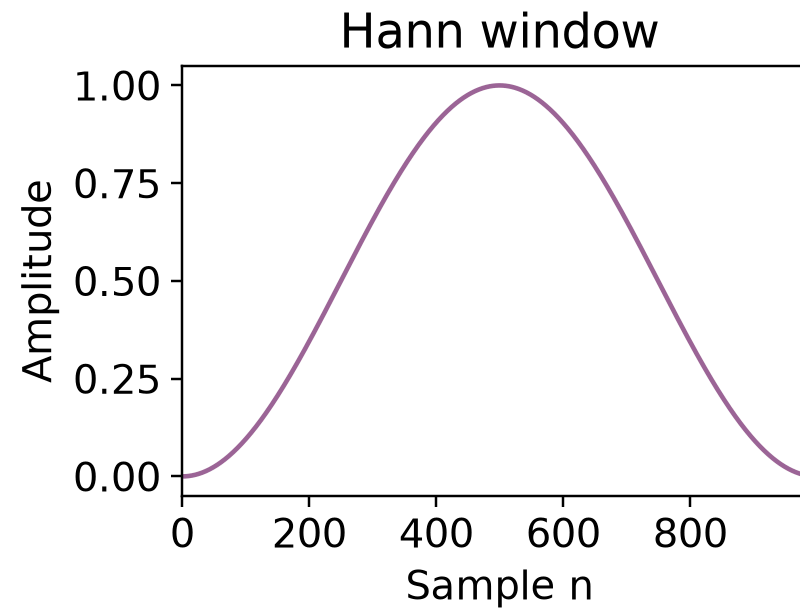
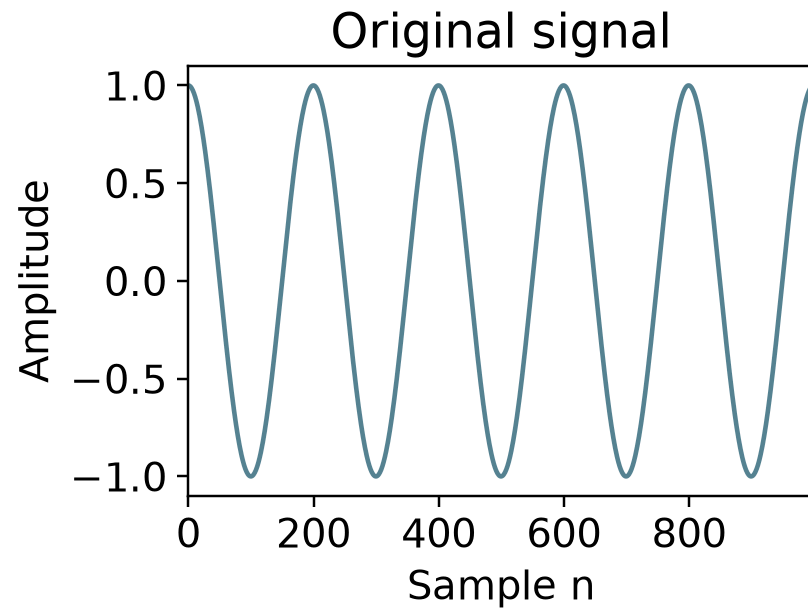


Two complete periods



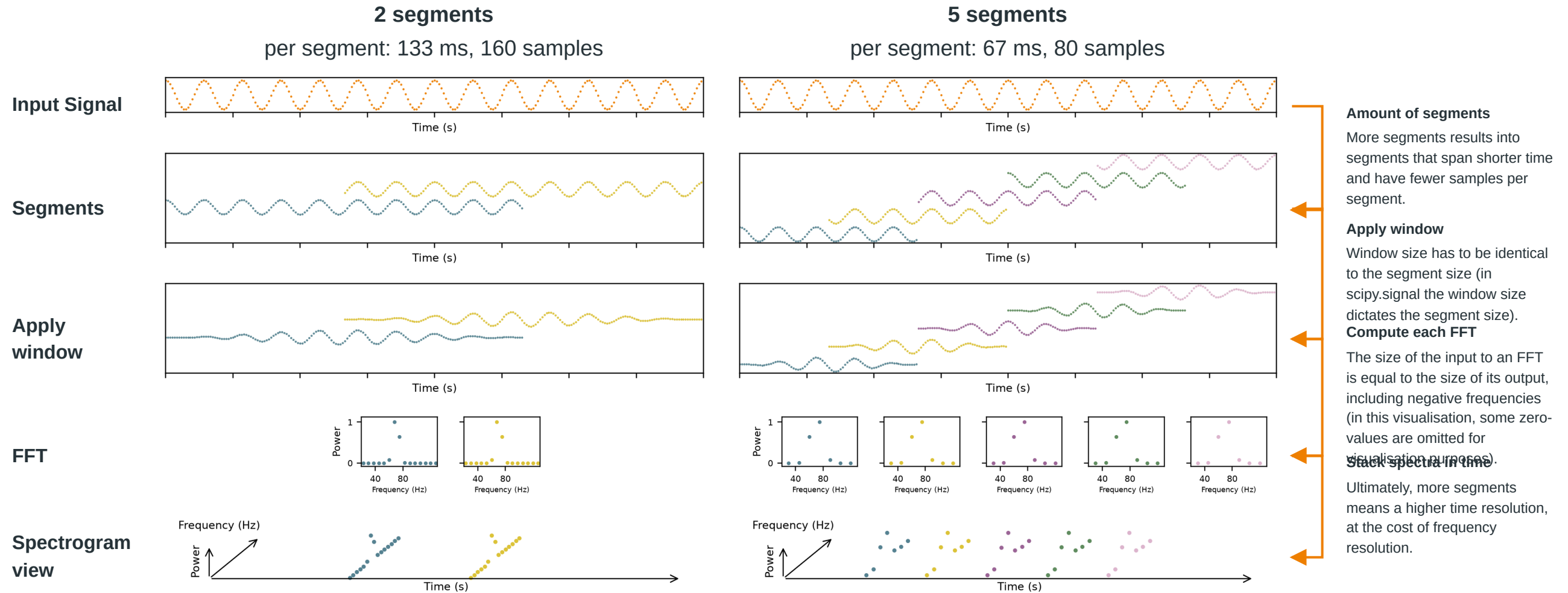
| Windowing

Windowing the signal

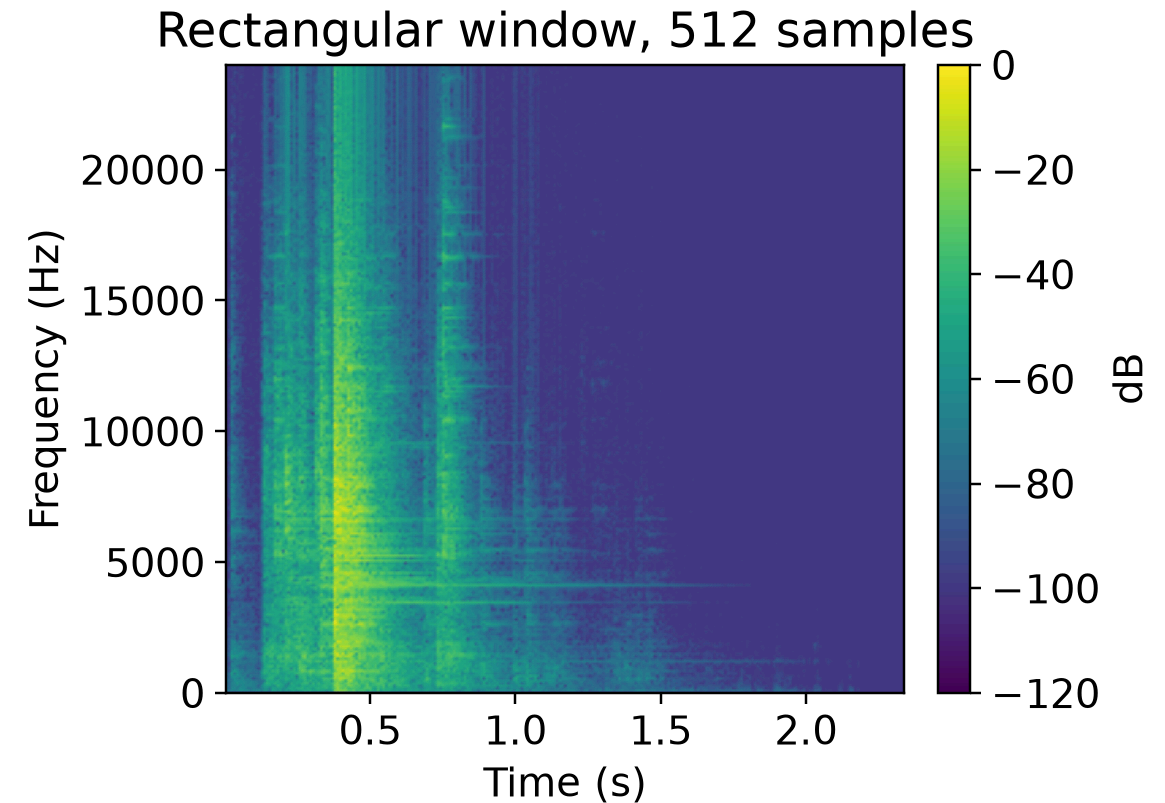
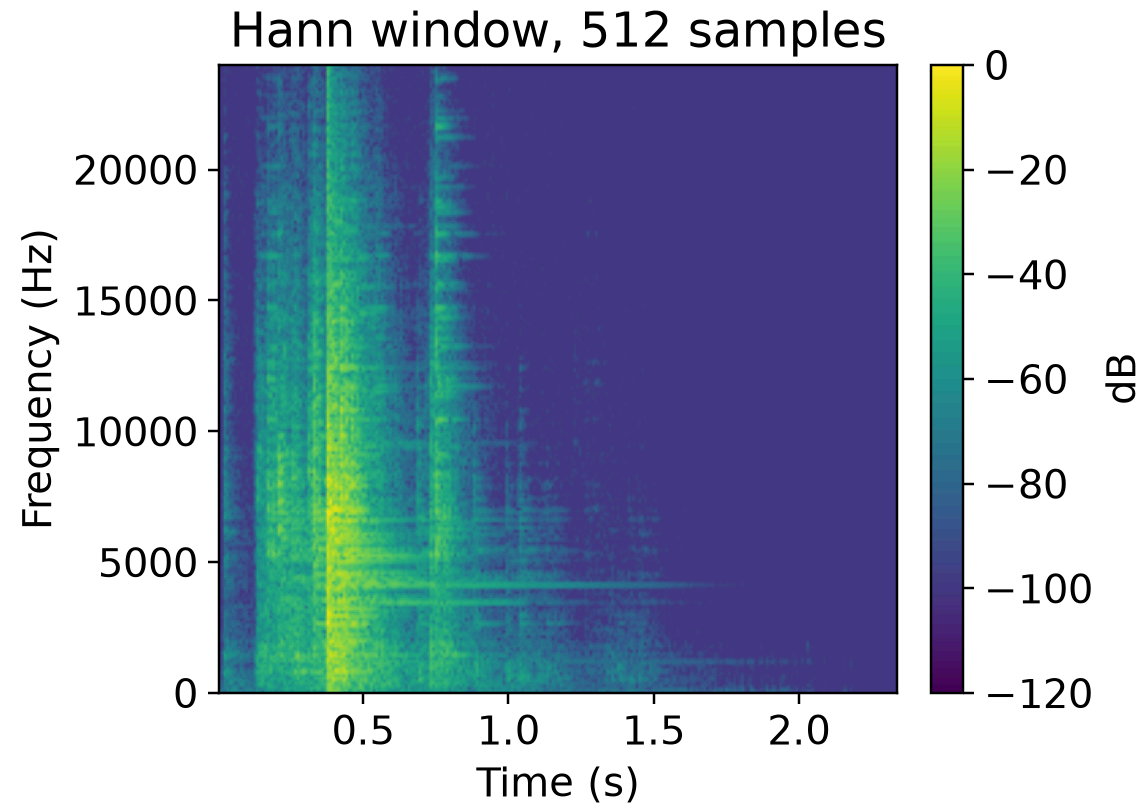


Multiply the signal segment by the window before taking its DFT.

The time-frequency uncertainty principle



Hann and rectangular windows



A Hann window reduces spectral leakage by smoothing segment edges, at the cost of broader frequency peaks.

| Exercises

Exercises!

`../exercises/DSP-Lecture-05-Exercises.ipynb`